

Association Between HPV Status and DNA Methylation in Head and Neck Squamous Cell Carcinoma

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INTRODUCTION

Head and neck squamous cell carcinoma (HNSCC) is a highly malignant disease that can arise in various locations within the head and neck region. While smoking and alcohol consumption are general causes of HNSCC, human papillomavirus (HPV) has recently been identified as a significant factor associated with HNSCC.¹

In our analysis, we found that the 5-year survival rate for HPV-positive patients is 0.71, compared to 0.59 for HPV-negative patients. HPV-positive HNSCC is associated with a better prognosis and an increased sensitivity to chemotherapy and chemoradiation.² Global methylation levels are highly related to HPV status, which may lead to distinct biological functions and clinical outcomes.¹

Our overall goal is to identify the genes with methylation levels associated with HPV status in HNSCC.

AIM

- Explore the association between HPV status and global DNA methylation levels in HNSCC.
- Identify specific genes with differential methylation patterns between HPV-positive and HPV-negative tumors.
- Determine if these differentially methylated genes can serve as biomarkers for prognosis and potential therapeutic targets.

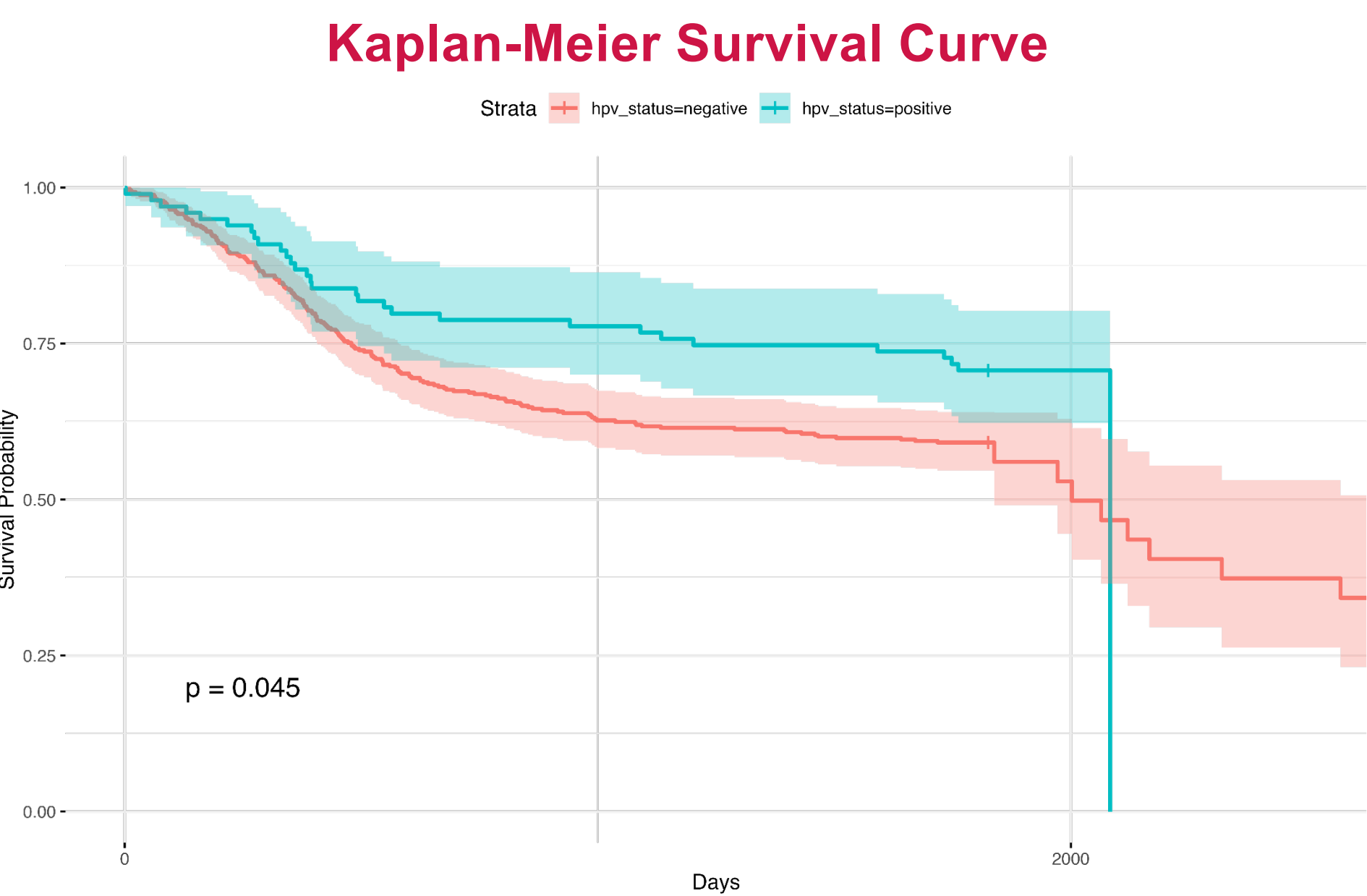


Figure 1. Overall survival rates of HPV-positive and HPV-negative HNSCC patients.

METHODS

Collection:

- Data Source: The Center Genome Atlas (TCGA) database

Preprocessing:

- Clinical Data: Excluded patients with indeterminate HPV status and non-head and neck tissue sites.
- Methylation Data: Removed redundant variables, CpG islands with missing beta-value, and addressed duplicate patient IDs by selecting solid tumor sample.

Patient Demographics

		HPV(+)	HPV(-)
Age	Mean	57.62	61.62
	Median	57.00	62.00
	Range	35 - 82	19 - 90
Sex	Female	12 (12.12%)	129 (30.21%)
	Male	87 (87.88%)	298 (69.79%)
Subdivision	alveolar ridge	5 (5.05%)	13 (3.04%)
	base of tongue	16 (16.16%)	11 (2.58%)
	buccal mucosa	0 (0.00%)	23 (5.39%)
	floor of mouth	4 (4.04%)	59 (13.82%)
	hard palate	1 (1.01%)	6 (1.41%)
	hypopharynx	5 (5.05%)	5 (1.17%)
	larynx	6 (6.06%)	110 (25.76%)
	lip	0 (0.00%)	3 (0.70%)
	oral cavity	8 (8.08%)	65 (15.22%)
	oral tongue	14 (14.14%)	118 (27.63%)
	oropharynx	2 (2.02%)	7 (1.64%)
	tonsil	38 (38.38%)	7 (1.64%)

Table 1. Demographics of patients with HPV-positive and HPV-negative HNSCC, including age, sex, and anatomical subdivisions.

Analysis:

- Calculated global methylation levels and compared them between HPV-positive and HPV-negative groups using box plots and Mann-Whitney tests.
- Filtered genes with low methylation levels and conducted t-tests to identify significant differences.
- Performed gene network/pathway analysis.

RESULTS

Global Methylation:

- HPV-positive patients exhibit significantly higher global methylation levels compared to HPV-negative patients ($p = 0.0368$, Mann-Whitney Test).

Differential Methylation:

- Identifies 93 genes with significant differences with thresholds of $|FC| > 1.3$ and $p < 0.05$.

Global Methylation Levels by HPV Groups

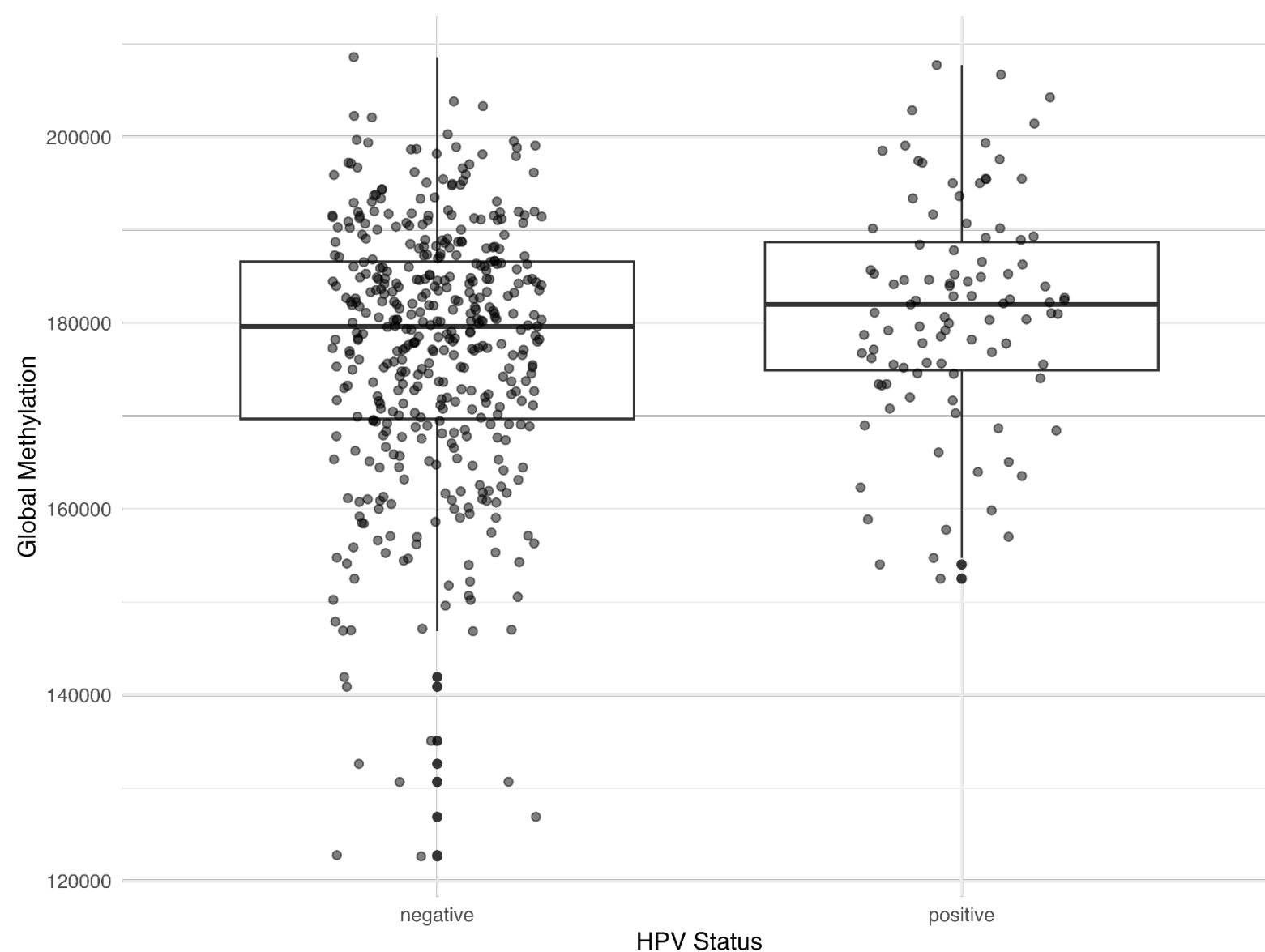


Figure 2. Global methylation comparison between HPV-positive and HPV-negative patients

Volcano Plot of Differential Methylation

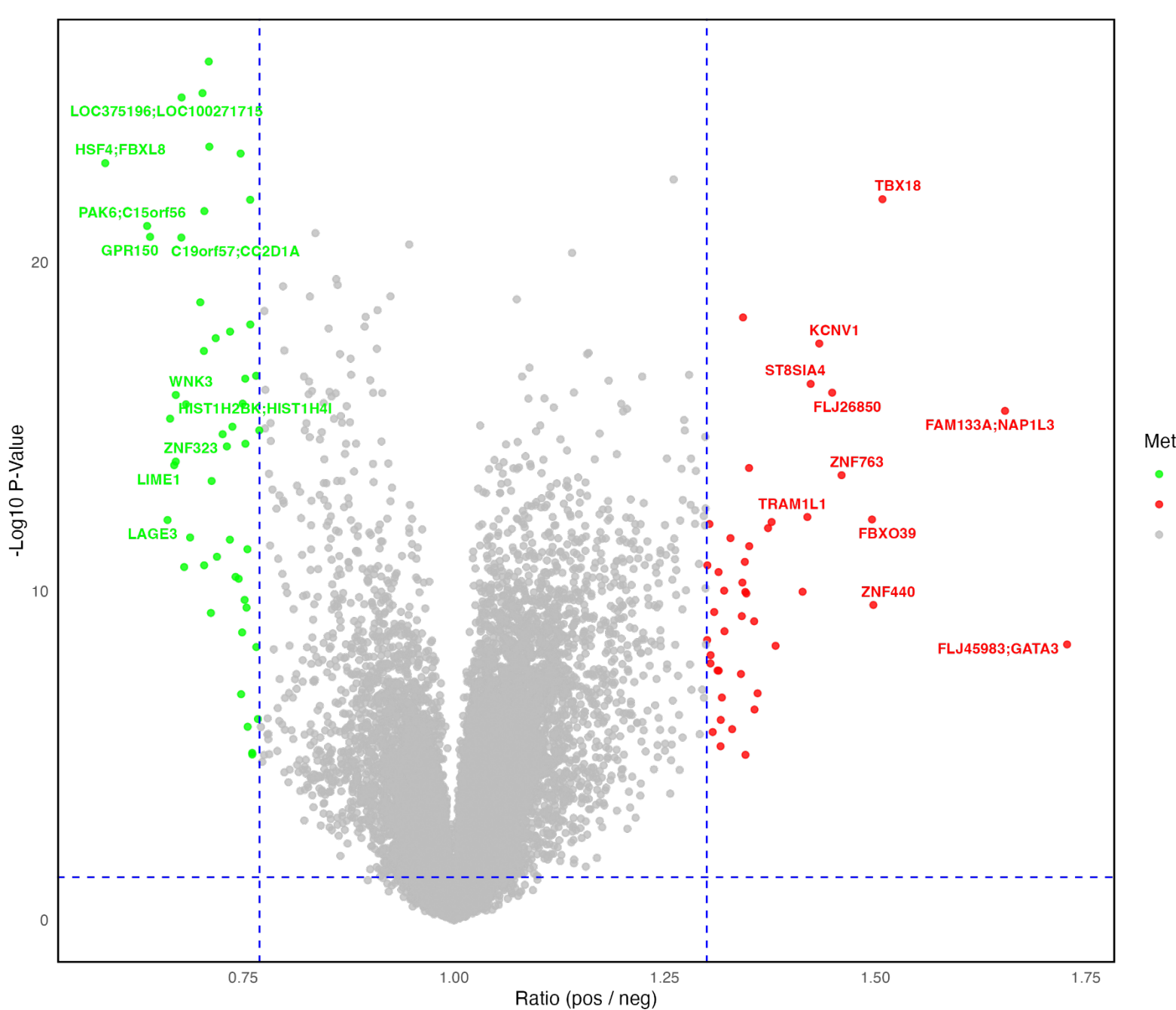


Figure 3. Differential methylation analysis of genes between HPV-positive and HPV-negative groups..

Heatmap of Gene Methylation by HPV Status

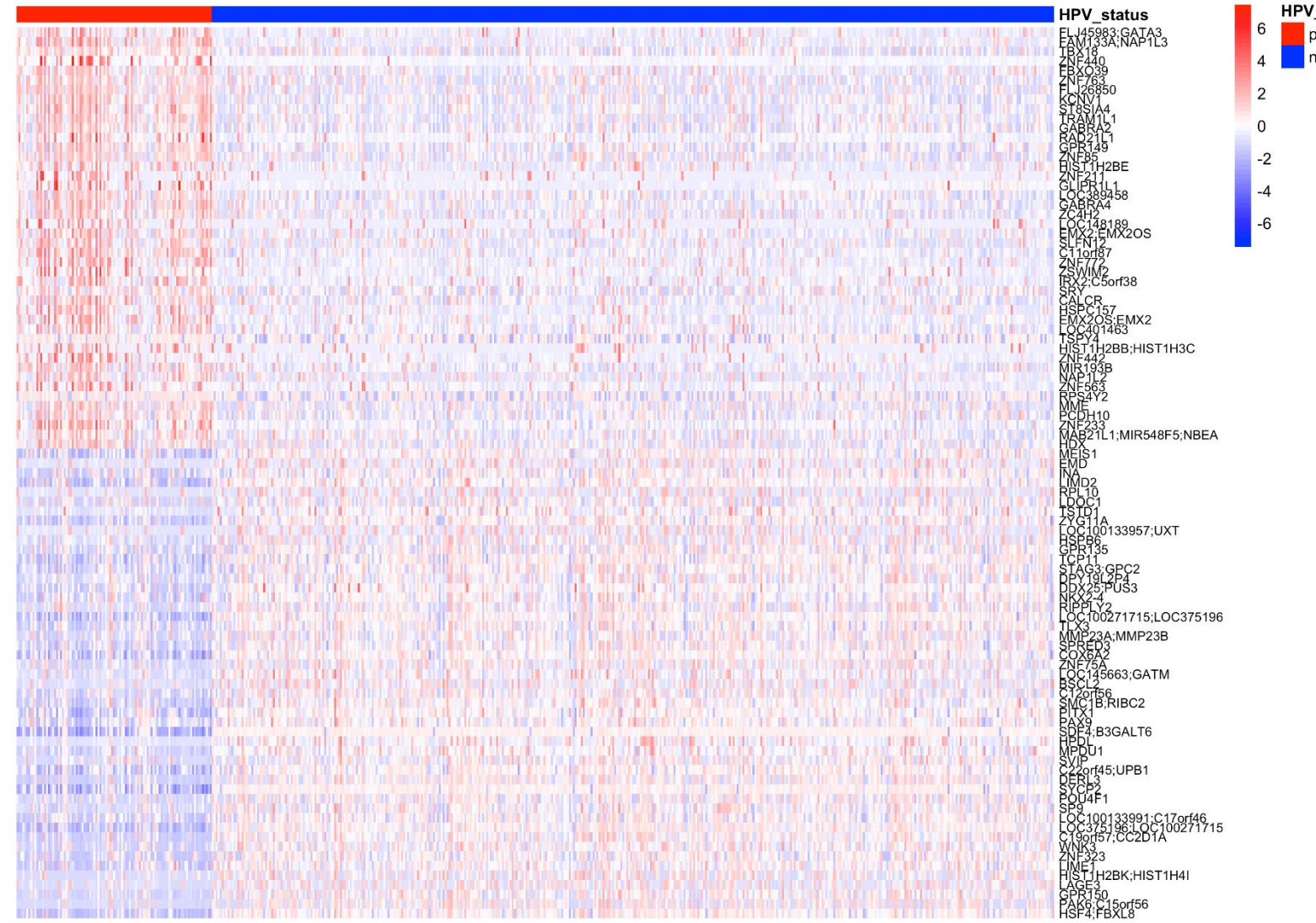


Figure 4. Methylation levels of significant genes between HPV-positive and HPV-negative groups. Red indicates higher methylation levels, and blue indicates lower methylation levels.

Gene Network:

- Key regulatory genes such as TBX18, GATA3, and PAX9 are identified. NOTCH1 and PI3K are involved in immune responses, indicating roles in tumor immune evasion.
- Highlights interactions with immunoglobulin pathways and miR-193, suggesting targets for immunotherapy and drug development in HPV-associated HNSCC.

Gene Interaction Network

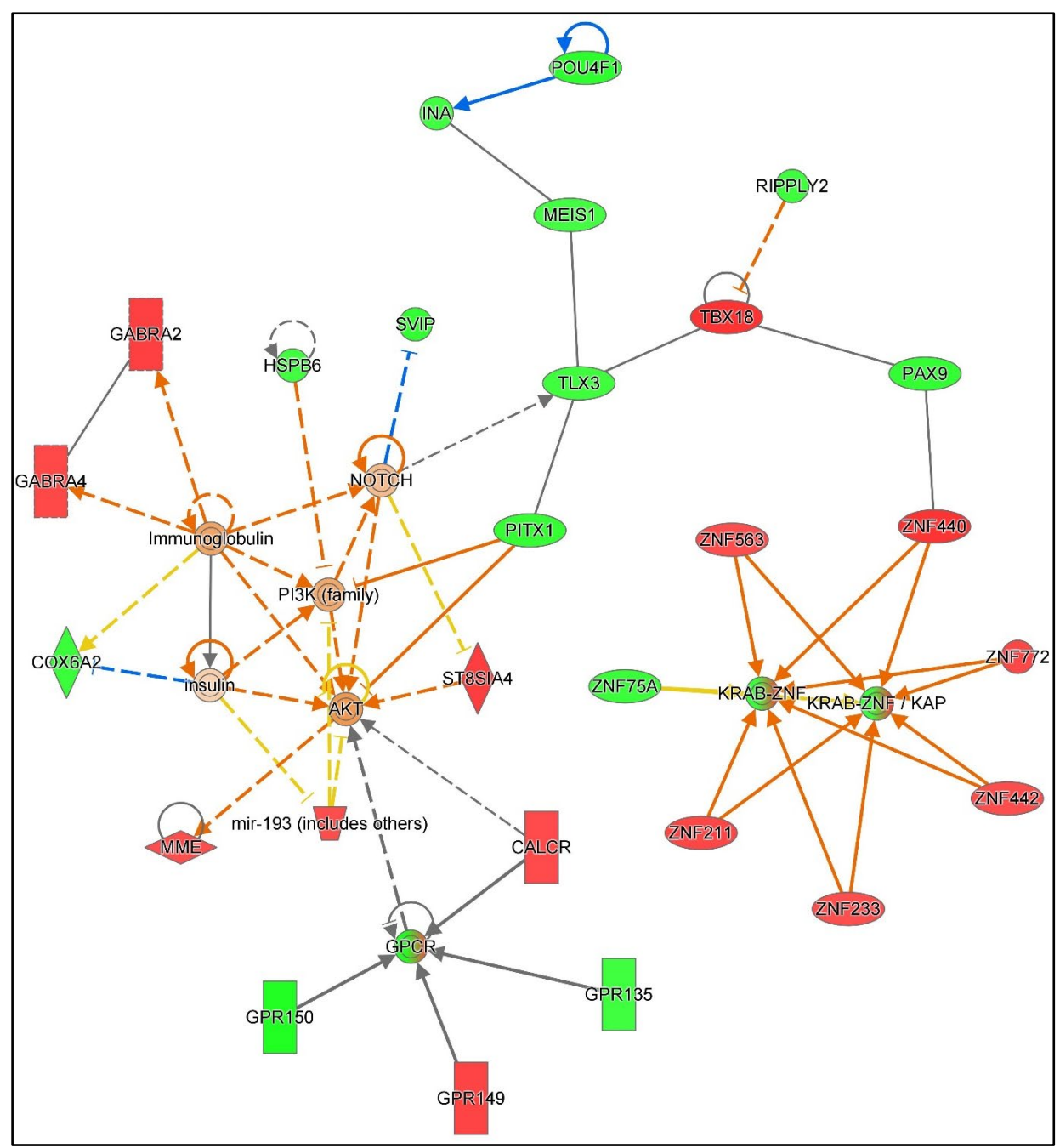


Figure 5. Interactions between differentially methylated genes in HPV-positive and HPV-negative patients. Highlights key pathways involved in immune response, cell cycle regulation, and DNA repair. Analysis using IPA software (www.qiagen.com/ingenuity).

CONCLUSIONS

- HPV status is significantly associated with differential DNA methylation patterns in HNSCC. HPV-positive tumors exhibit higher global methylation levels, key regulatory genes such as TBX18, GATA3, and PAX9 are identified.
- Future studies should focus on validating these findings in larger cohorts, exploring the potential therapeutic targets, and examining specific anatomical subdivisions such as the tonsil and throat.

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